

On the inevitability of a homicidal ancestor: a genealogical–probabilistic argument

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Abstract

We start from an existential question: *what is the probability that neither I nor any of my ancestors ever caused the death of another human being?* We formalise this as a problem of compound probability over the ascending genealogical tree. We show that, because of the exponential growth in the number of ancestors, this probability collapses to values indistinguishable from zero within a few dozen generations, even under extremely conservative assumptions about the prevalence of lethal violence. Moreover, invoking the *identical ancestors point* (Rohde et al., 2004), the argument ceases to be probabilistic and becomes essentially deductive: beyond that horizon, the set of ancestors of any living person coincides with the entire global breeding population of that era, so that the existence of a single killer in that pool suffices. We support the result with a Monte Carlo simulation, and we examine isolated populations and the relaxation of the independence assumption. We conclude that it is practically impossible not to descend from at least one killer.

1 Introduction

Many people intuit that their lineage could have been free of lethal violence: “not I, not my parents, not my grandparents, nobody back through time”. This note shows that such an intuition is mathematically untenable. The argument does not rest on assuming that humans are especially violent, but on a purely combinatorial property of kinship: the number of ancestors grows exponentially into the past, and a logical conjunction (“*no one* was a killer”) over an enormous number of individuals becomes, in practice, impossible to satisfy.

We combine three ingredients: (i) a naive model of the doubling of ancestors; (ii) the correction due to *pedigree collapse*; and (iii) the *identical ancestors point* (IAP), which turns the probabilistic result into a practical certainty.

2 Definitions and assumptions

Killer. We adopt a broad and deliberate definition: a person is a *killer* if they caused the death of another human being by any intentional or quasi-intentional cause (war, self-defence, intergroup conflict, etc.). This choice maximises the robustness of the argument and avoids the legal and moral ambiguities that depend on the era.

*Author’s note: this work arises from a personal inquiry driven by curiosity; it is offered as a rigorous and reproducible synthesis, in a scientific spirit, without claiming academic credentials.

Ancestor. We use the *genealogical* (lineal) sense: every parent, grandparent, great-grandparent, etc., regardless of whether or not they transmitted genetic material to the descendant. The distinction between genealogical and genetic ancestry is relevant and is discussed in Section 6 (Ralph and Coop, 2013).

Parameters. Let p be the probability that any given individual is a killer over their lifetime, and $q = 1 - p$. Let N be the number of generations considered into the past and $\ell \approx 28$ years the mean generation length (Fenner, 2005).

Independence assumption. In the analytic models we assume that the events “being a killer” are independent across individuals. This is a simplification (there is family and regional correlation); its effect is examined in Section 6 and relaxed in the simulation.

3 Probabilistic model

3.1 Naive model

The number of ancestors doubles each generation: 2 parents, 4 grandparents, $\dots$, 2^n at generation n . The total number of individuals in the lineage (including *ego*) up to generation N is

$$T(N) = \sum_{k=0}^N 2^k = 2^{N+1} - 1. \quad (1)$$

Under independence, the probability that *none* of them is a killer is

$$P_{\text{naive}}(p, N) = (1 - p)^{T(N)} = \exp[(2^{N+1} - 1) \ln(1 - p)]. \quad (2)$$

Since $T(N)$ grows exponentially, for any fixed $p > 0$ we have $P_{\text{naive}}(p, N) \rightarrow 0$ very quickly (Fig. 1).

3.2 Pedigree collapse

The naive model overcounts: the “slots” 2^n soon exceed the human population of the era, so a single individual occupies many positions of the tree (*pedigree collapse*). The number of *distinct* ancestors at generation n is bounded by the breeding population $\Pi(n)$:

$$D(n) = \min(2^n, \Pi(n)). \quad (3)$$

Saturation occurs around $n^* = \lceil \log_2 \Pi \rceil$. For $\Pi \sim 10^8$, $n^* \approx 27$ (~ 750 years); for $\Pi \sim 8 \times 10^9$, $n^* \approx 33$ (~ 920 years) (Fig. 2). Far from rescuing the intuition, collapse *worsens* it: beyond n^* the ancestors of *ego* are practically the entire population of their era.

3.3 Identical ancestors point (the strong argument)

Chang (1999) showed that, in a well-mixed population of size N_e , one need only go back of order $1.77 \log_2 N_e$ generations for every individual of that time (with descendants today) to be an ancestor of *all* the present population. Rohde et al. (2004) estimated, by simulation with realistic migration, that the most recent common ancestor (MRCA) of *Homo sapiens* lived ~ 2000 – 5000 years ago, and that the *identical ancestors point* (IAP) —the moment beyond which every person of that time with descendants is an ancestor of all the living— lies, depending on the mixing assumptions, approximately 5000 – 15000 years ago (with bounds of 1000 – 15000 years in more conservative estimates).

The consequence is decisive. Let $\mathcal{E}$ be the global breeding population at an era prior to the IAP, of size $|\mathcal{E}|$. Beyond the IAP, *the set of ancestors of any living person is exactly $\mathcal{E}$* . Therefore, the probability of descending from at least one killer is

$$P_{\text{desc}}(p) = 1 - (1 - p)^{|\mathcal{E}|}. \quad (4)$$

With $|\mathcal{E}|$ of order millions (or billions), an infinitesimal p suffices for $P_{\text{desc}}(p) \approx 1$ (Table 2). The statement ceases to be probabilistic: if at least one killer *existed* in $\mathcal{E}$ —something empirically certain (Section 4)—then *every* living human descends from a killer.

4 Empirical parameters

To fix p we use as an anchor the phylogenetic analysis of Gómez et al. (2016): the proportion of human deaths *predicted* by comparative methods to be due to interpersonal violence is $\approx 2\%$, a value consistent with that inferred for the evolutionary ancestor of primates and with that observed in prehistoric bands and tribes (and exceeded in many historical periods). In an approximately stationary population, each violent death implies at least one perpetrator, so the fraction of people who ever killed is of the same order of magnitude: $p \sim 10^{-2}$. Considering war and intergroup conflict over millennia, p could be substantially higher. As a precaution, we explore a wide range $p \in [10^{-4}, 0.15]$ and show that the conclusion holds even at its lower end.

As a bound on the number of individuals involved, we use the estimate that $\sim 1.17 \times 10^{11}$ human beings have ever been born (Kaneda and Haub, 2022) (other estimates give $\sim 1.08 \times 10^{11}$); the number of *distinct* ancestors of any person is bounded by this figure. For the size of the breeding population in each era we use the long-run demographic series of Ritchie et al. (2023).

5 Results

Naive model. Table 1 and Fig. 1 show the collapse of $P_{\text{naive}}(p, N)$. Already at $N = 15$ generations (~ 420 years) the probability is astronomically small except for unrealistically low p .

Table 1: $P_{\text{naive}}(p, N)$: probability that *no* ancestor (naive model) is a killer.

N	individuals	$p = 0.001$	$p = 0.01$	$p = 0.05$	$p = 0.15$
5	63	9.4×10^{-1}	5.3×10^{-1}	4.0×10^{-2}	3.6×10^{-5}
10	2 047	1.3×10^{-1}	1.2×10^{-9}	2.5×10^{-46}	3.3×10^{-145}
15	65 535	3.3×10^{-29}	9.0×10^{-287}	≈ 0	≈ 0
20	2 097 151	≈ 0	≈ 0	≈ 0	≈ 0

IAP argument. Table 2 evaluates Eq. (4) for a breeding pool of $|\mathcal{E}| = 5 \times 10^6$. Even with $p = 10^{-6}$ (one in a million), the probability of descending from a killer exceeds 0.99; for $p \geq 10^{-5}$ it is, numerically, equal to 1.

Monte Carlo simulation. We build a population of $M = 5\,000$ individuals per generation over 30 generations with random mating; each individual is a killer independently with probability p , and we propagate the indicator “has at least one homicidal ancestor” (average of 20 replicates). Even with $p = 10^{-4}$, the fraction of the present population with a homicidal

Table 2: $P_{\text{desc}}(p) = 1 - (1 - p)^{|\mathcal{E}|}$ with $|\mathcal{E}| = 5 \times 10^6$.

p	$P_{\text{desc}}(p)$
10^{-6}	0.9933
10^{-5}	1.0000
10^{-4}	1.0000
10^{-3}	1.0000
10^{-2}	1.0000

ancestor reaches practically 1 by generation 20–29; with $p = 10^{-2}$, by generation 10 (Table 3). The simulation relaxes the infinite-population assumption of the naive model and confirms the result.

Table 3: Fraction of the population with ≥ 1 homicidal ancestor (Monte Carlo, $M = 5\,000$, 20 replicates).

generation	$p = 10^{-4}$	$p = 10^{-3}$	$p = 10^{-2}$
5	0.006	0.060	0.460
10	0.155	0.826	1.000
15	0.764	1.000	1.000
20	0.998	1.000	1.000
29	1.000	1.000	1.000

6 Discussion and limitations

Independence. The independence hypothesis is false: the propensity to violence correlates within families and regions, and warlike events affect many at once. However, positive correlation tends to *reinforce* the conclusion (it concentrates killers in some branches, but a single branch suffices to break the conjunction), and the random-mating simulation already dispenses with the infinite population.

Heritability and correlation. The propensity to violence has a hereditary and culturally transmitted component (Anholt and Mackay, 2012), which introduces correlation between the status of parents and children. To assess its effect we extend the simulation with a transmission parameter $h \in [0, 1]$, $p_{\text{child}} = p + h(f - p)$, where f is the fraction of the two parents who were killers; the model *preserves the marginal prevalence* p , so it compares the independent case ($h = 0$) with the clustered one ($h > 0$) at equal mean violence. The simulated prevalence stays at ≈ 0.010 for all h (Table 4), and even with $h = 0.9$ the fraction with a homicidal ancestor converges to 1, crossing 0.999 at generation 10 versus generation 9 in the independent case (Fig. 4). That is, correlation barely delays the phenomenon by one generation and does not change the conclusion: the thesis is robust to the independence assumption.

Genealogical vs. genetic. The argument is *genealogical*. An ancestor at the IAP may have left no genetic material in *ego* (Ralph and Coop, 2013). This does not affect the claim about descent, which is genealogical, not genetic, in nature.

Sensitivity to the definition. Under a strict legal definition of “murder” (unlawful homicide), p would be smaller; even so, Tables 2 and 3 show that the result holds for p as low as

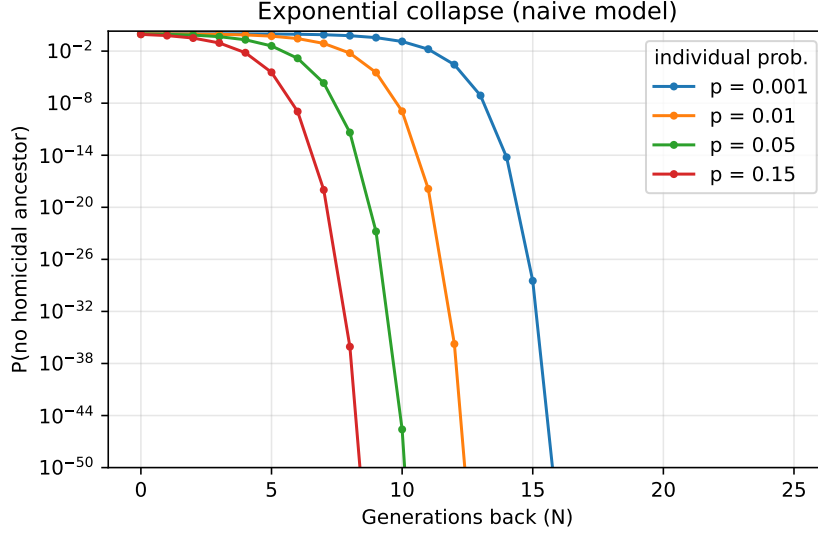


Figure 1: Exponential collapse of $P_{\text{naive}}(p, N)$ (logarithmic scale) according to the naive model, Eq. (2).

Table 4: Robustness to independence (Monte Carlo, $p = 0.01$, $M = 2000$, 40 replicates). h is the heritability/transmission; the marginal prevalence is preserved.

h	prevalence	gen. with frac. ≥ 0.999	fraction (gen. 29)
0.0	0.010	9	1.000
0.3	0.010	9	1.000
0.6	0.010	10	1.000
0.9	0.010	10	1.000

10^{-4} .

Population mixing. The model of Rohde et al. (2004) assumes global mixing, mitigated in reality by geographic barriers; this delays the global IAP but does not eliminate it, and the *regional* IAP is even more recent.

Isolated populations. One might ask whether an isolated population (an island, a remote valley, a settlement founded by a few colonists) could escape the conclusion. The analysis shows the opposite, in three ways. (i) *Founder effect*: in a closed, well-mixed population of effective size N_e , the IAP is reached in $\approx 1.77 \log_2 N_e$ generations (Chang, 1999); for $N_e = 50$ this is ~ 10 generations (~ 280 years), versus ~ 35 for $N_e = 10^6$ (Table 5, Fig. 5). Isolation therefore *accelerates* the collapse of ancestry: on a small island everyone soon shares the same pool of ancestors. (ii) *Heterogeneity of p* : some isolated societies show very high lethal violence (Keeley, 1996) —the (disputed) case of the Yanomamö reports high fractions of men who have killed (Chagnon, 1988)— while others are notably peaceful; isolation increases the *variance* of p , not necessarily its mean. (iii) *Not an escape route*: even in the limiting case of a small peaceful isolate, where the closed pool alone does not guarantee a killer (e.g. $N_e = 50$, $p = 10^{-3}$ gives $P_{\text{desc}} \approx 0.05$; Table 5), the *founders* in turn descend from the mainland population, whose ancestry connects with the global IAP (Section 4), which does contain killers. Isolation only adds a recent genealogical “cap”; it does not sever deep ancestry, so certainty is restored through the founders themselves.

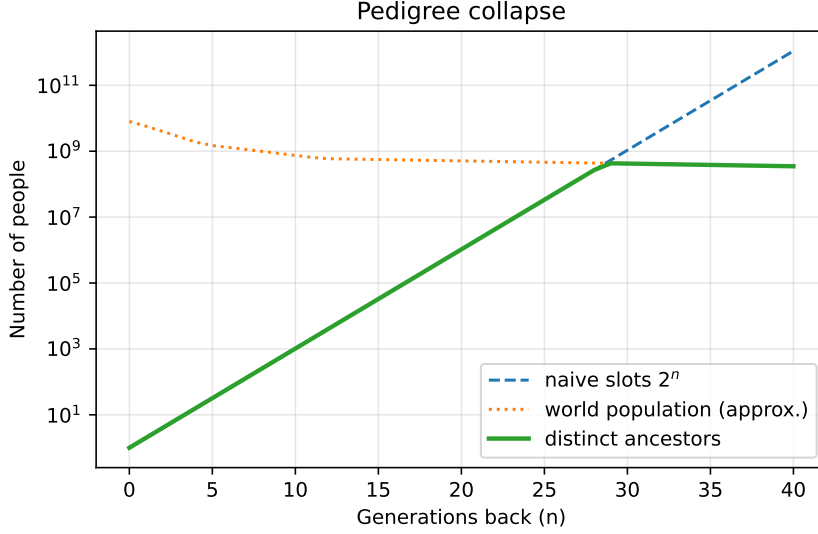


Figure 2: Pedigree collapse: the naive slots 2^n exceed the population, so the distinct ancestors $D(n)$ saturate, Eq. (3).

Table 5: Isolated (closed) populations: generation of the IAP ($\approx 1.77 \log_2 N_e$) and probability of descending from a killer *within the isolated pool*, $P_{\text{desc}} = 1 - (1 - p)^{N_e}$.

N_e	IAP (gen.)	$P_{\text{desc}} (p=10^{-3})$	$P_{\text{desc}} (p=10^{-2})$
50	10.0	0.049	0.395
200	13.5	0.181	0.866
1000	17.6	0.632	1.000
10 000	23.5	1.000	1.000
10^6	35.3	1.000	1.000

Positioning and novelty. This work does not claim a new mathematical result: it is an expository *synthesis* that combines already-known results —the recent common ancestry of Chang (1999) and Rohde et al. (2004), and the prevalence of lethal violence of Gómez et al. (2016)— to answer a concrete existential question quantitatively. Its contribution is pedagogical and integrative: it makes explicit, with reproducible code and sensitivity analysis, a corollary that usually remains implicit in the genetic-genealogy literature.

7 Conclusion

The probability of having no homicidal ancestor is, under any realistic parametrisation, indistinguishable from zero. The exponential growth of ancestors and pedigree collapse suffice to make it minuscule within a few generations; the identical ancestors point turns it into a practical certainty of an almost deductive character. Therefore, *it is essentially impossible for a living human not to descend from at least one killer*. The existential intuition that motivated this note is thus rigorously refuted.

On the nature of the data

This work *does not collect primary data*. It is worth distinguishing three kinds of figures. (i) The *empirical values* (prevalence of lethal violence, IAP horizon, population size, generation

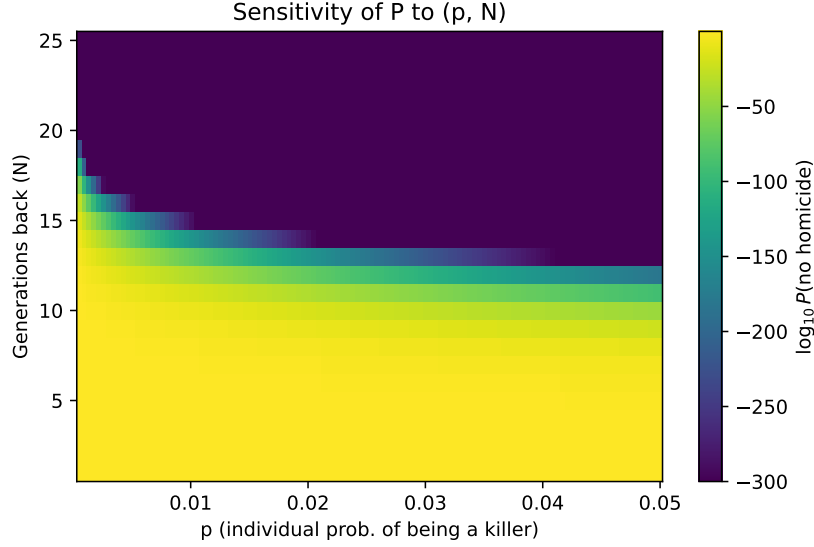


Figure 3: Sensitivity of $\log_{10} P_{\text{naive}}$ to (p, N) . The dark region (probability ≈ 0) dominates except for very small p and very small N .

length, etc.) are *secondary*: they come from external, peer-reviewed sources, cited in the text and traced one by one—with DOI and verbatim quotation—in the file `EVIDENCE.md` of the repository. (ii) The *numerical results* (tables, figures and simulations) do not depend on the author’s authority but are *reproducible*: they are obtained by running the open code with fixed seeds. (iii) The *model and its derivations* are verifiable by mathematical inspection. In addition, the parameters are taken as orders of magnitude and the conclusion is robust over wide ranges (Sections 4 and 6), so it does not depend on any single value. Consequently, verification of any claim is referred to its cited source or to re-running the code, not to the author’s testimony.

Reproducibility

All numerical results, tables and figures are generated by the code in the `code/` directory (Python; see `requirements.txt`). The Monte Carlo simulation uses fixed seeds, so its results are deterministic and reproducible. This document is compiled with Tectonic (`tectonic main-en.tex`).

Data and code availability

The code, derived data and sources of this article are available in the project repository (<https://github.com/JuanoLemos/paper-antepasado-homicida>) and are permanently archived in Zenodo with DOI 10.5281/zenodo.20806914. The traceability of each claim to its primary source is documented in the file `EVIDENCE.md`. The code is distributed under the MIT licence and the text and figures under CC-BY-4.0.

Competing interests

The author declares no competing interests.

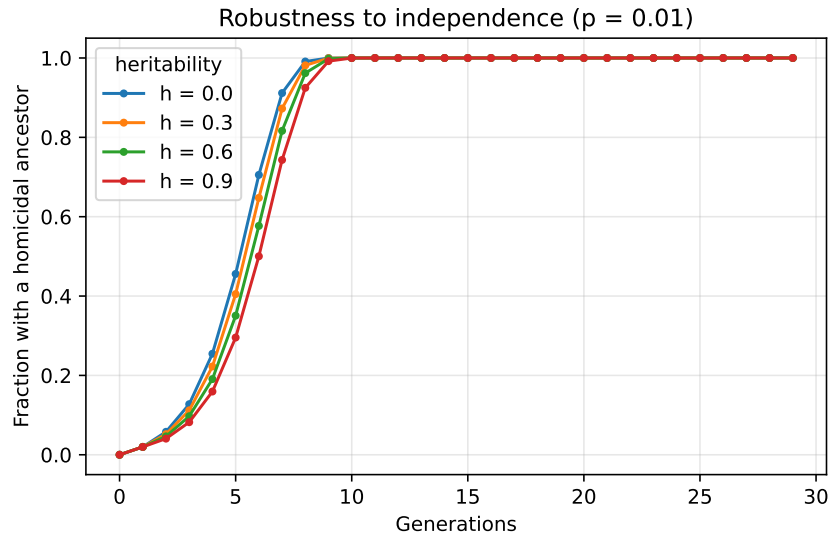


Figure 4: Robustness to independence: fraction with a homicidal ancestor per generation for different levels of heritability/correlation h , at fixed marginal prevalence $p = 0.01$. The curves are almost indistinguishable.

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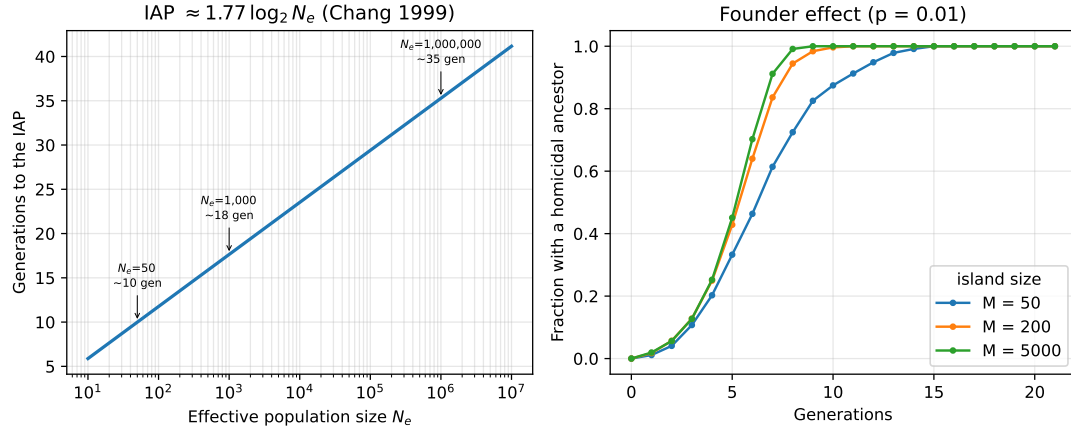


Figure 5: Isolated populations. Left: generation of the IAP versus the effective size N_e ($\approx 1.77 \log_2 N_e$). Right: fraction with a homicidal ancestor per generation in closed populations of different size M (founder effect, $p = 0.01$, Monte Carlo).

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